Lecture 23 (11/9)

Regularization

Review of material since midterm 2

1) Using interactions to data where the effect of one variable depends on the others
2) Using residual plots to diagnose a model
3) Building nonlinear models with features, polynomial regression
4) Overfitting and bias-variance tradeoff as model complexity increases

$$
\begin{aligned}
E_D[\hat{\sigma}^2_{\text{test}}] &= E_X\left[\text{MSE}_{\hat{y}(X,D)}\right] + \sigma^2_\epsilon \\
&= \underbrace{E_X\left[\text{var}_D(\hat{y}(X,D))\right]}_{\text{variance}} + \underbrace{E_X\left[(E_D[\hat{y}(X,D)] - f(X))^2\right]}_{\text{bias}^2} + \sigma^2_\epsilon
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Today:

1) Review and practice for unit 5 material
2) Begin discussing regularization, a technique to prevent overfitting

Practice questions: Conceptual

Question

Consider a model $\hat{y} = f(x)$ which predicts the cost of an individual's car insurance based on some attributes $x = (x_1, x_2, x_3, \dots)$. For example, x_1 might be number of previous crashes. Discuss the implications of bias and variance, both from the driver's and the insurance company's perspectives.

Practice questions: Technical

Question

A cancer treatment can use any combination of three drugs A, B and C, where $X_A = 1$ if drug A is used and $X_A = 0$ if not (and the same for X_B and X_C). The response Y is the reduction in tumor size (in mm). The model with all interactions is

$$Y = \beta_A X_A + \beta_B X_B + \beta_C X_C + J_{AB} X_A X_B + J_{AC} X_A X_C + J_{BC} X_B X_C + J_{ABC} X_A X_B X_C + \epsilon$$

with

$$\beta_A = 4, \quad \beta_B = 3, \quad \beta_C = 2,$$
$$J_{AB} = -2, \quad J_{AC} = 1, \quad J_{BC} = -4, \quad J_{ABC} = 2.$$

If drug A is unavailable, what is the best treatment?

Practice questions: Technical

Question

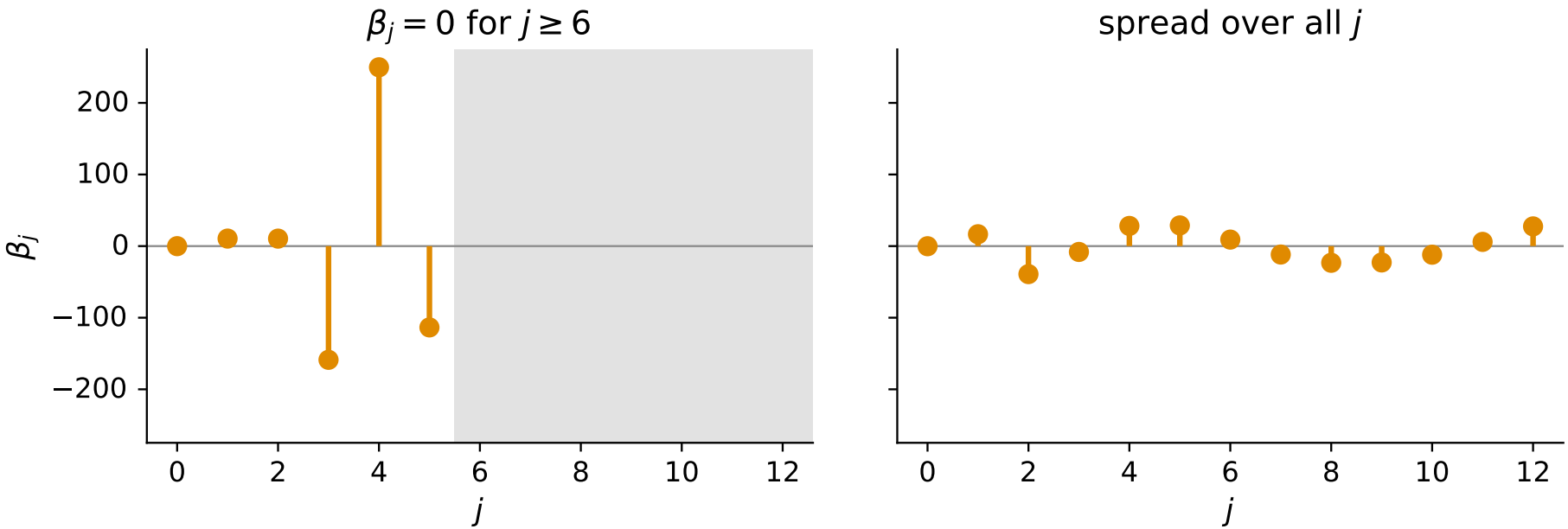
A degree-K polynomial is fit to $N_{\text{train}} = 100$ training points. The training error is $\hat{\sigma}^2_{\text{train}} = 0.5$ and $\hat{\sigma}^2_Y = 2$ for the training Y values. What is R^2 on the training points?

Question

1) Describe how $\hat{\sigma}^2_{\text{train}}$ and $\hat{\sigma}^2_{\text{test}}$ change as K is increased to N_{train}
2) If the model has high variance and low bias in the bias-variance decomposition of $E_D[\hat{\sigma}^2_{\text{test}}]$ does $\hat{y}(x, D^{\text{train}})$ have a large standard error?
3) Can $E_D[\hat{\sigma}^2_{\text{test}}]$ be smaller than σ^2_ϵ?

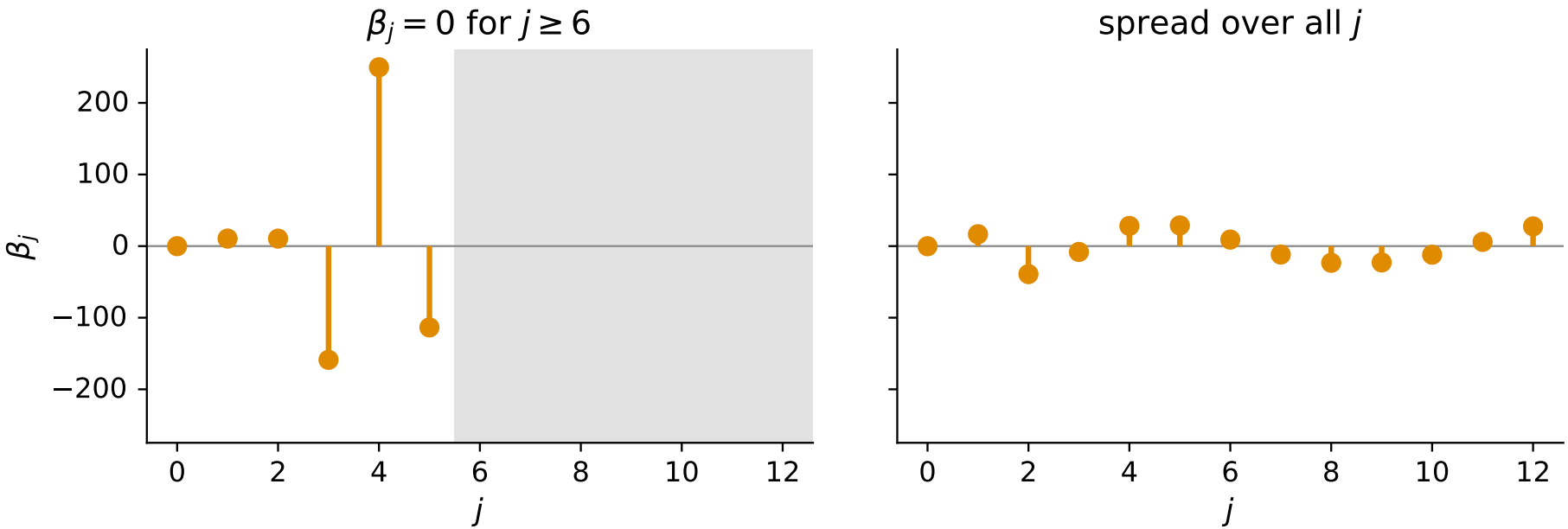
A hard cutoff, or coefficients that shrink?

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots$$



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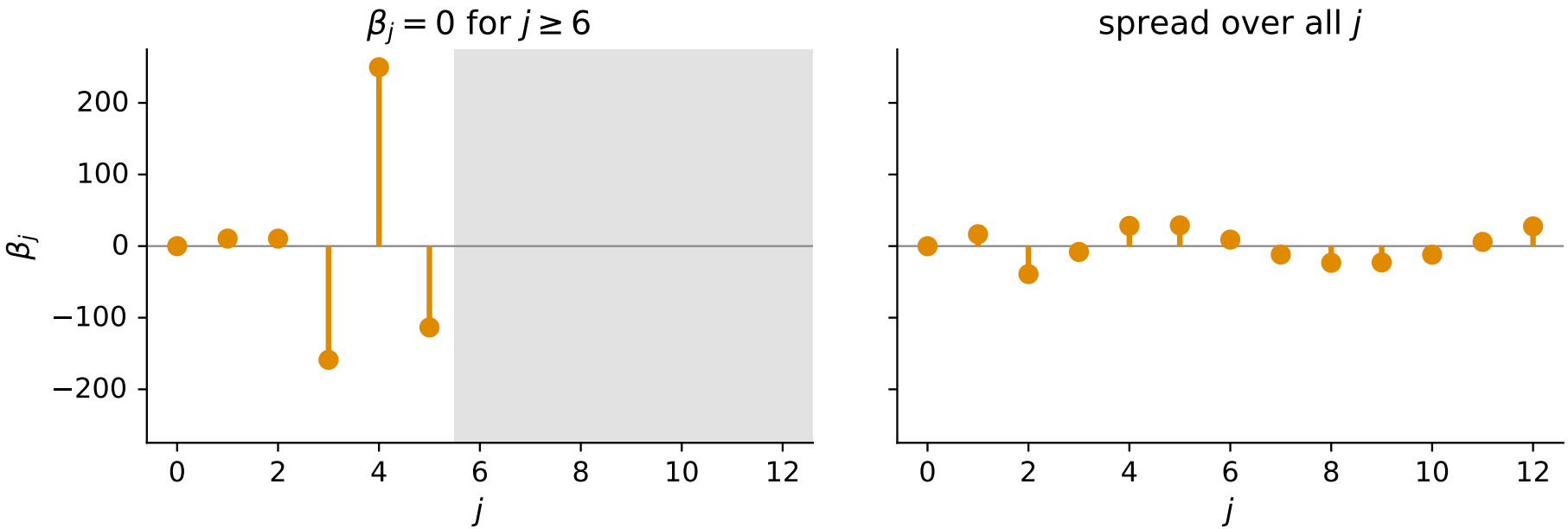
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1) When is having more, but smaller, coefficients preferable?
2) How do we implement this?

Optimization view of fitting a model

Consider the model

$$Y = \mu + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2), \qquad N = N_{\text{train}}.$$

We could also write $\mu = \beta_0$ to use regression notation. When we fit this model to data, we are just estimating the mean, and we know that an unbiased estimator is

$$\hat{\mu} = \overline{Y} = \frac{1}{N} \sum_{i=1}^{N} Y_i.$$

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In the least-squares view, we interpret this estimator as the solution of an optimization problem. It minimizes

$$L(\mu) = \frac{\text{SSR}}{N} = \frac{1}{N} \sum_{i=1}^{N} (\mu - Y_i)^2 = \hat{\sigma}^2_{\text{train}},$$

where $L(\mu)$ is called the loss function.

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Regularization

The idea of regularization is to force μ to be smaller by add a penalty to the loss for larger μ:

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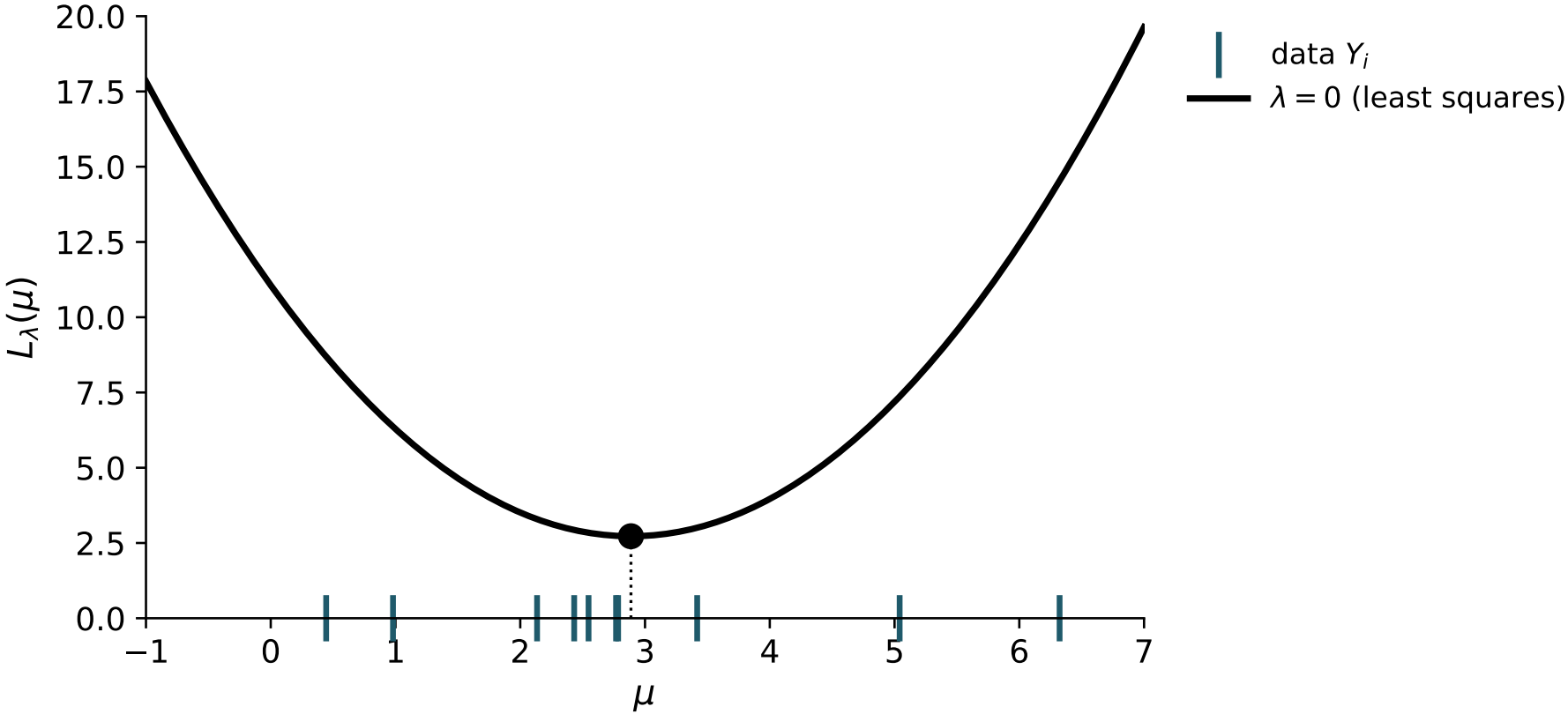
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As $\lambda \to \infty$ the estimate goes to 0. The larger λ, the smaller $\hat{\mu}^\lambda$.

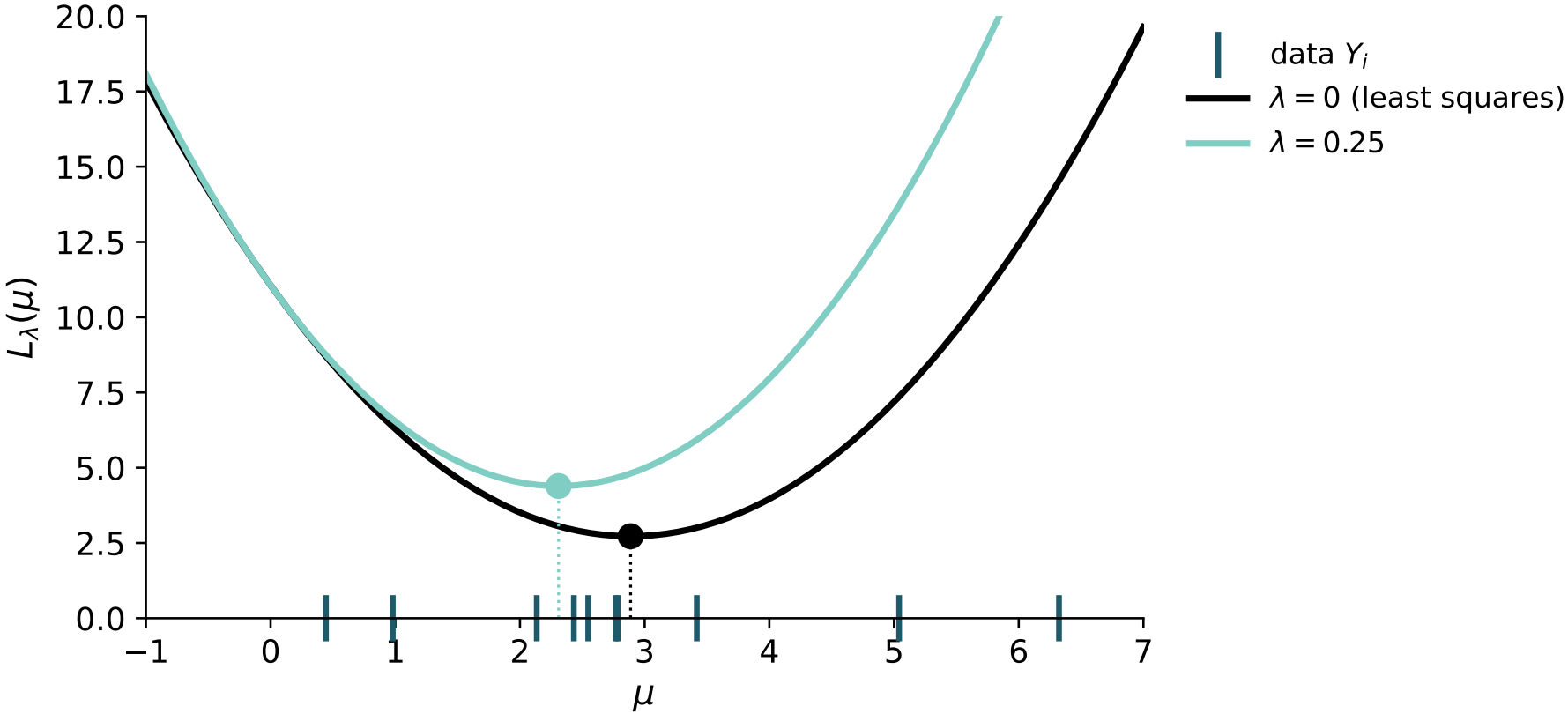
Loss for different λ



$$L_\lambda(\mu) = (1+\lambda)\left(\mu - \frac{\overline{Y}}{1+\lambda}\right)^2 + \hat{\sigma}_Y^2 + \frac{\lambda\,\overline{Y}^2}{1+\lambda}$$

Derive this: drill "Completing the square for the ridge loss of a mean", unit 6 notes §6.3.

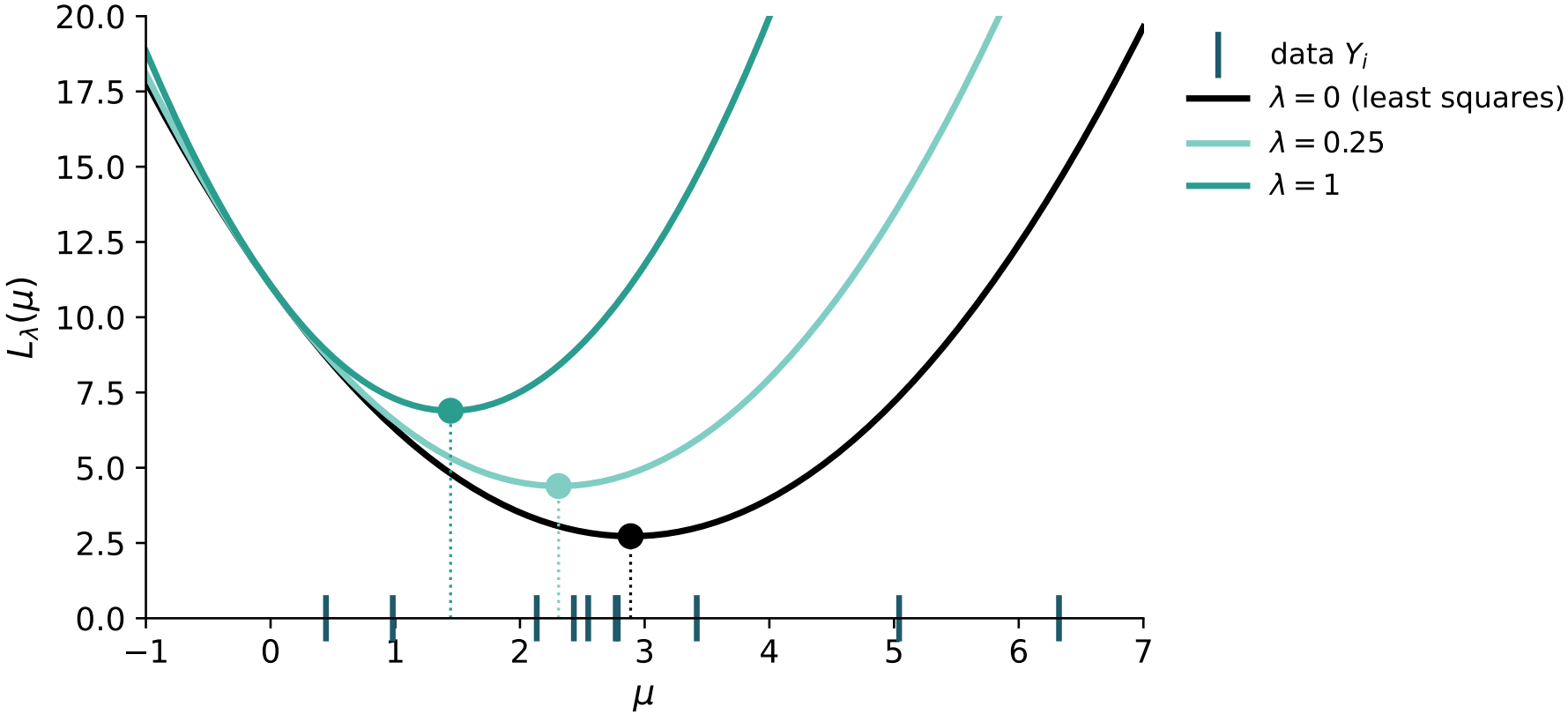
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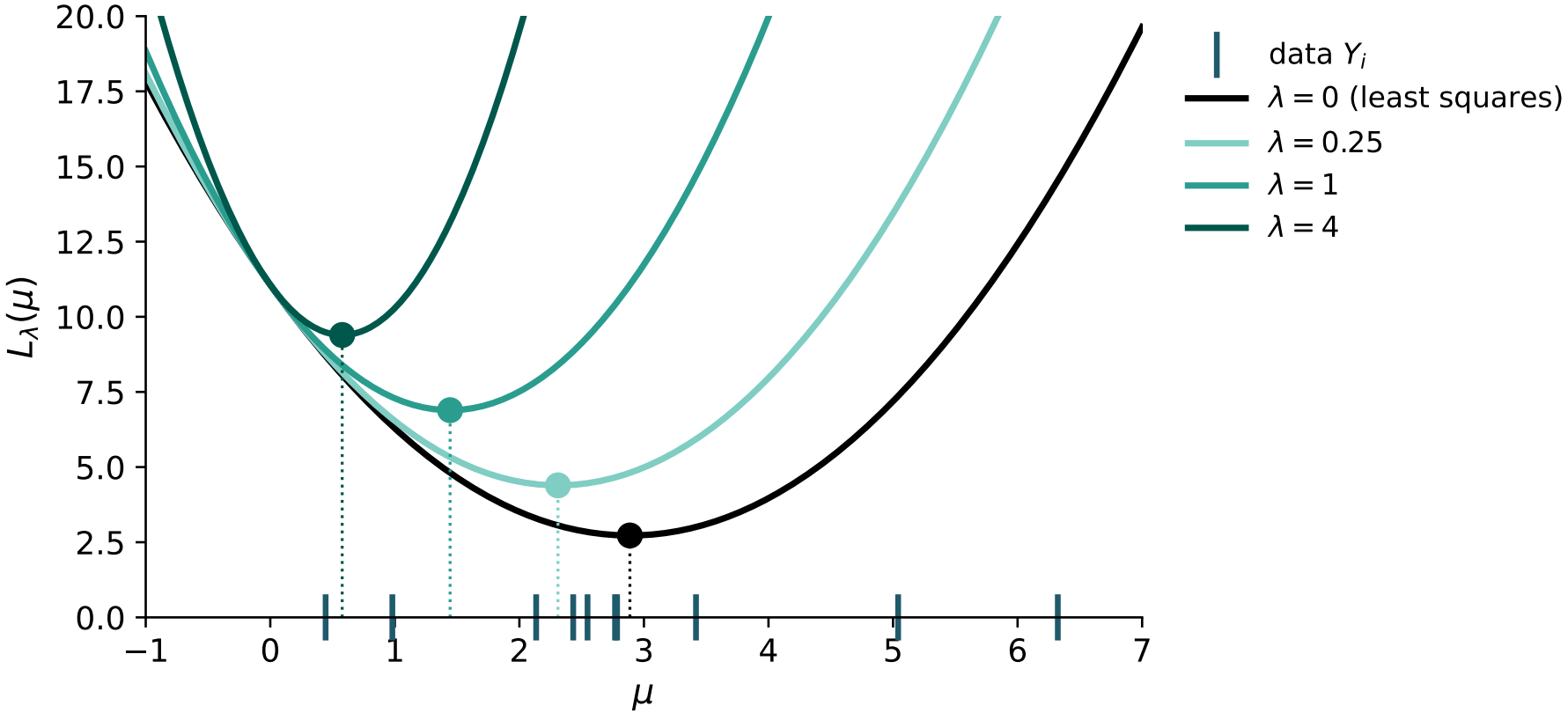
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Regularization for a linear model

Consider the model

$$Y = \beta_1 X + \epsilon, \quad \epsilon \sim \mathrm{Normal}(0, \sigma_\epsilon^2),$$

with $E[X] = 0$ and no intercept. The least-squares estimate is

$$\hat{\beta}_1 = \frac{\sum_i X_i Y_i}{\sum_i X_i^2}, \qquad \text{and we write} \quad \hat{\sigma}_X^2 = \frac{1}{N} \sum_i X_i^2.$$

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As $\lambda \to \infty$ the estimate goes to 0. The more spread out the X_i (larger $\hat{\sigma}_X^2$), the less a given λ shrinks it.

Regularization for a general regression model

least squares, degree K

$\hat{\beta}$ minimizes $\quad L(\beta) = \hat{\sigma}^2_{\text{train}} = \dfrac{1}{N_{\text{train}}} \displaystyle\sum_{i=1}^{N_{\text{train}}} \left(\hat{y}(X_i^{\text{train}}, D^{\text{train}}) - Y_i^{\text{train}}\right)^2$

each x^j is either in the model ($j \leq K$) or not

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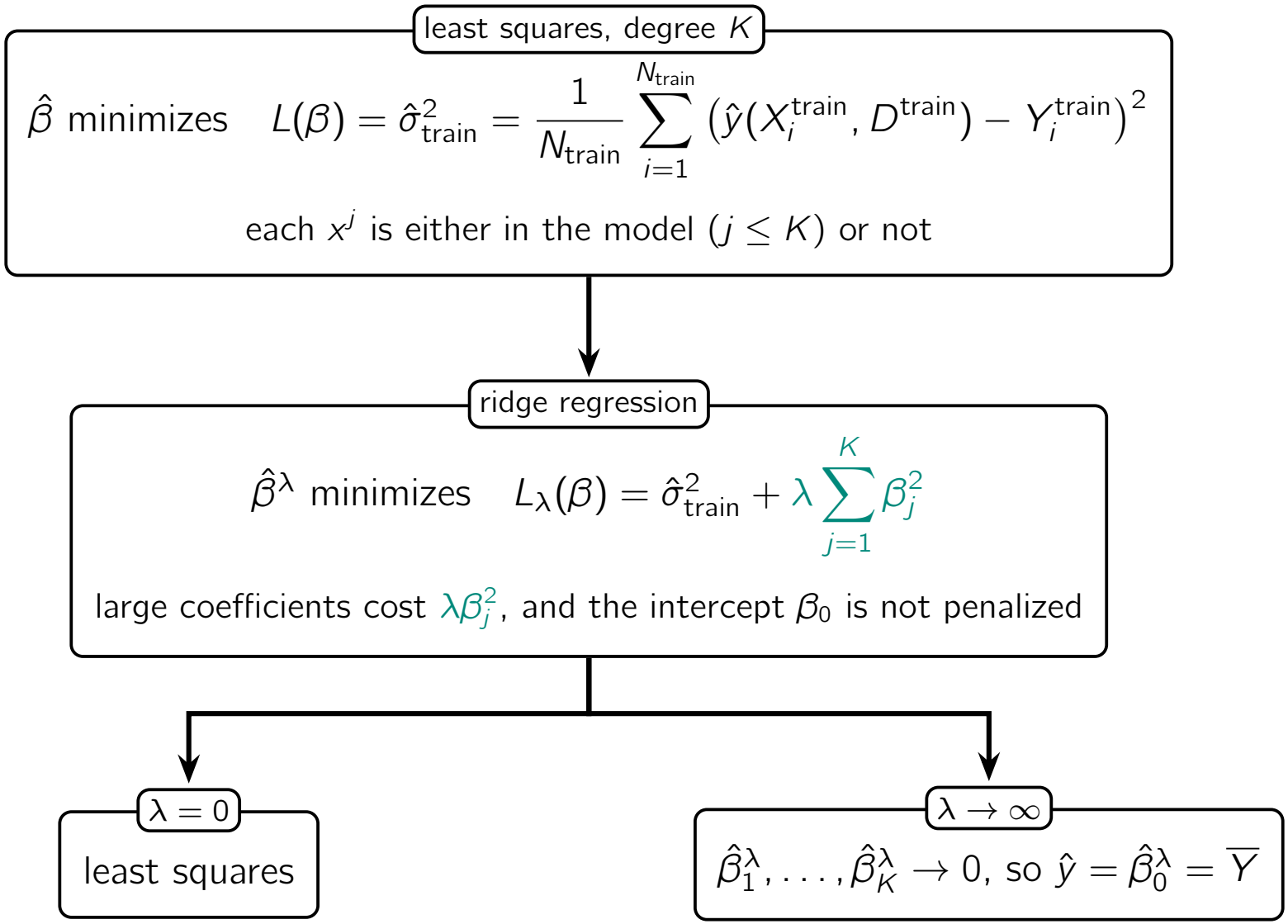
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ridge regression

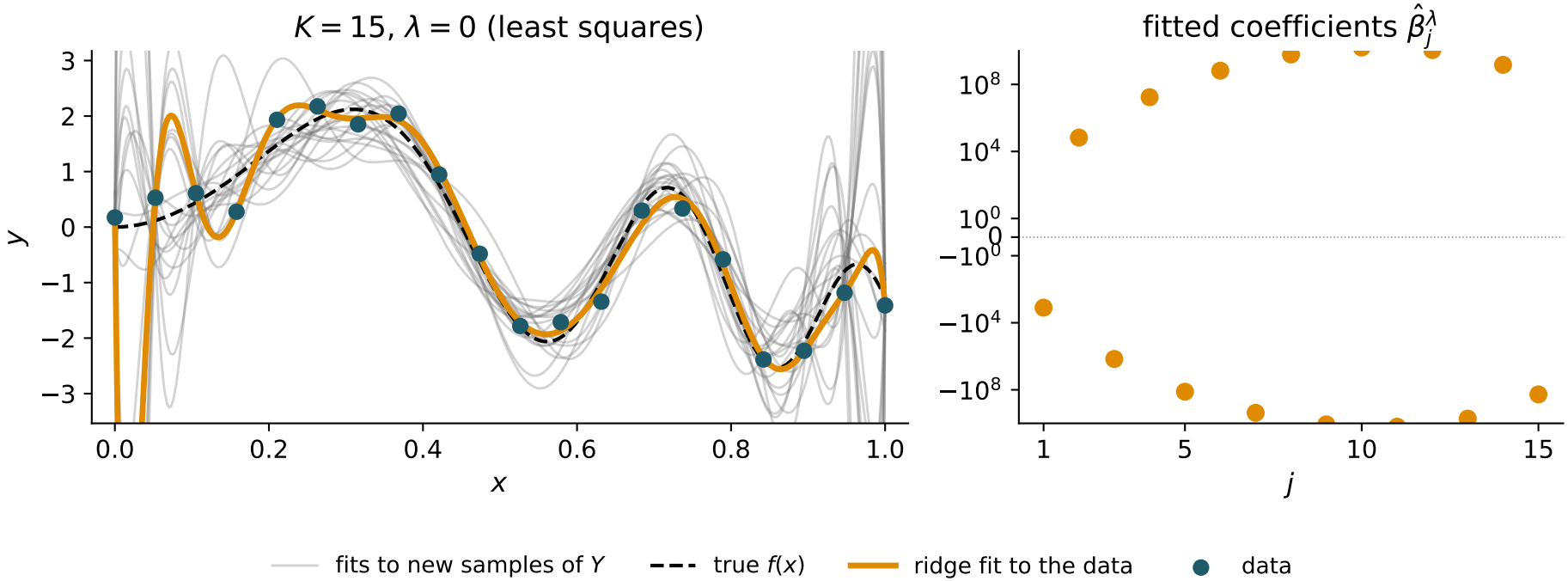
$\hat{\beta}^\lambda$ minimizes $\quad L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \displaystyle\sum_{j=1}^{K} \beta_j^2$

large coefficients cost $\lambda\beta_j^2$, and the intercept β_0 is not penalized

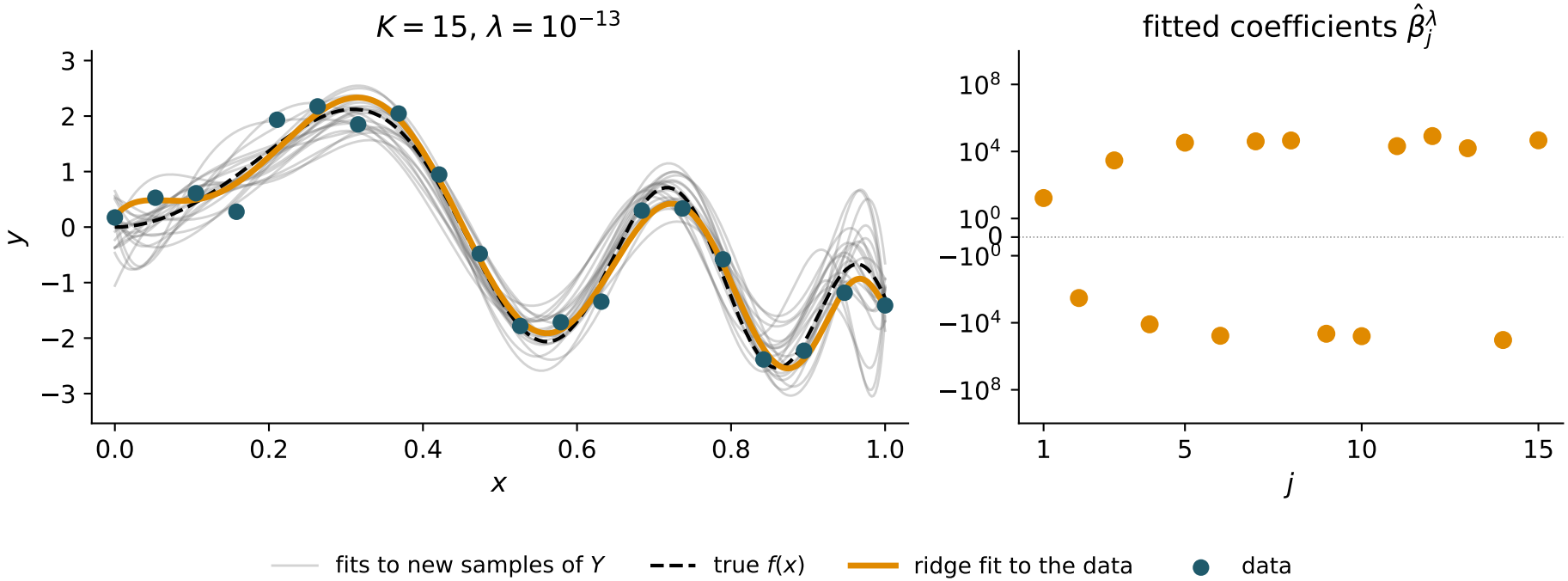
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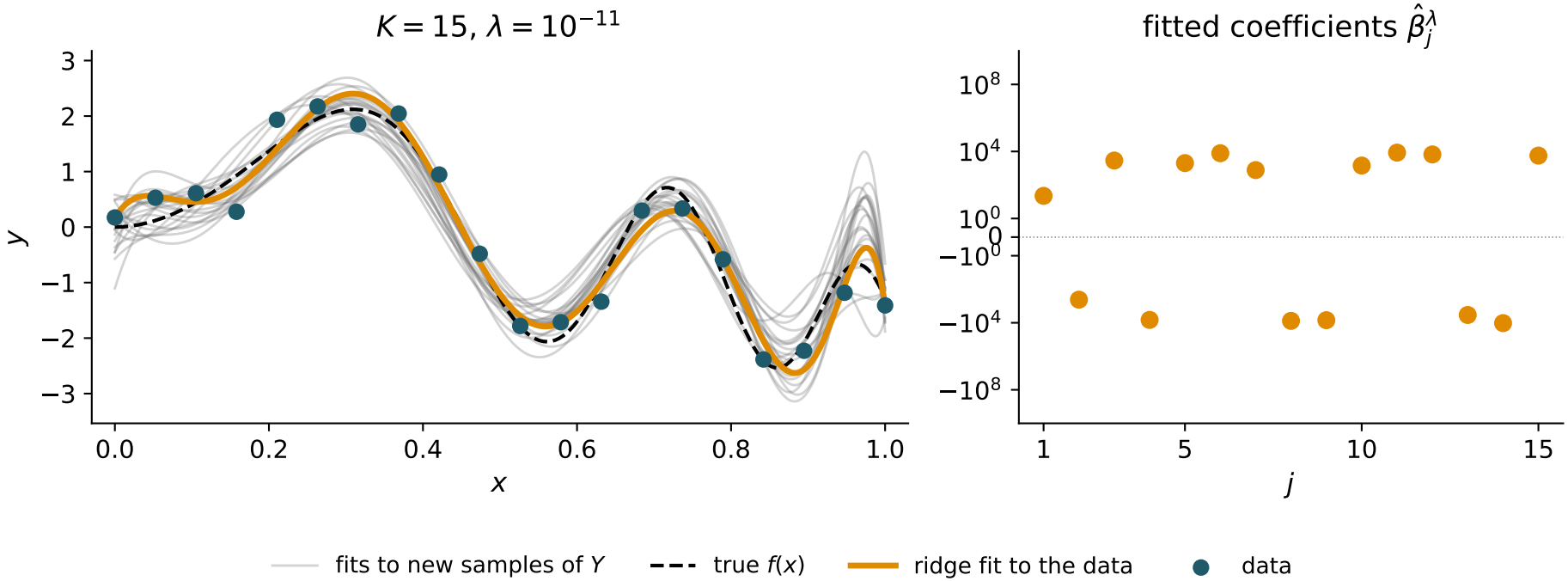
The effect of λ



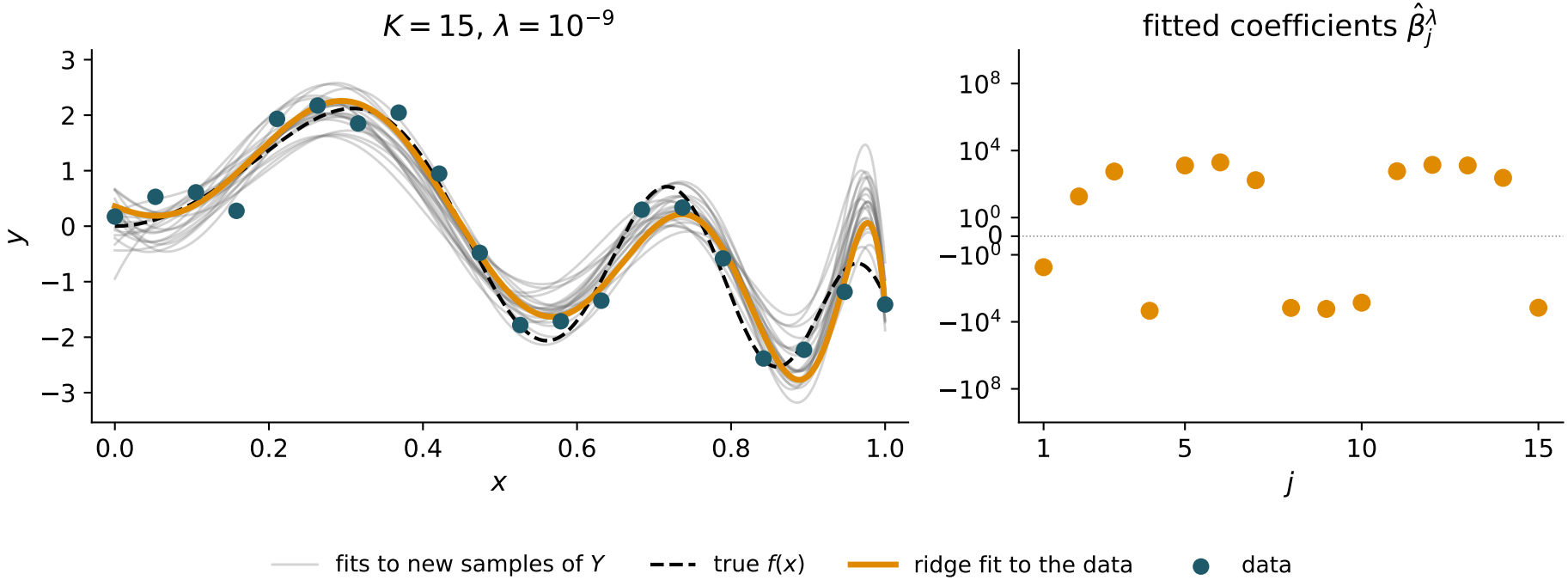
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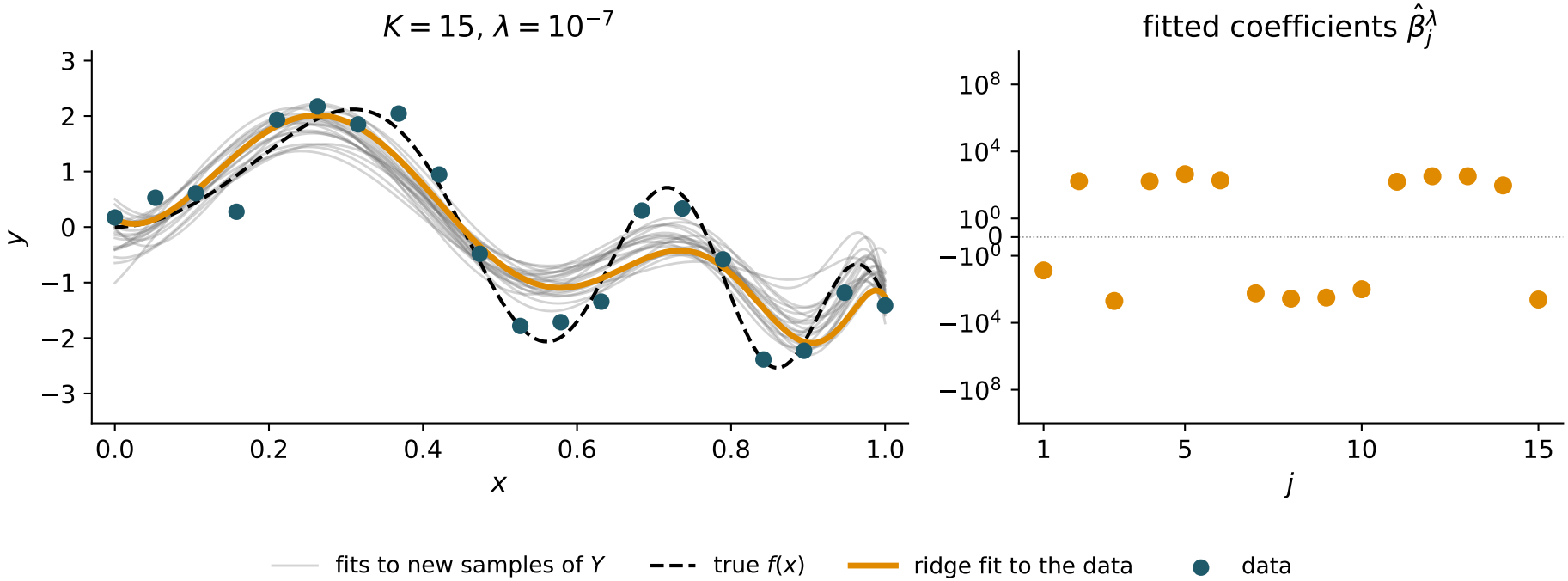
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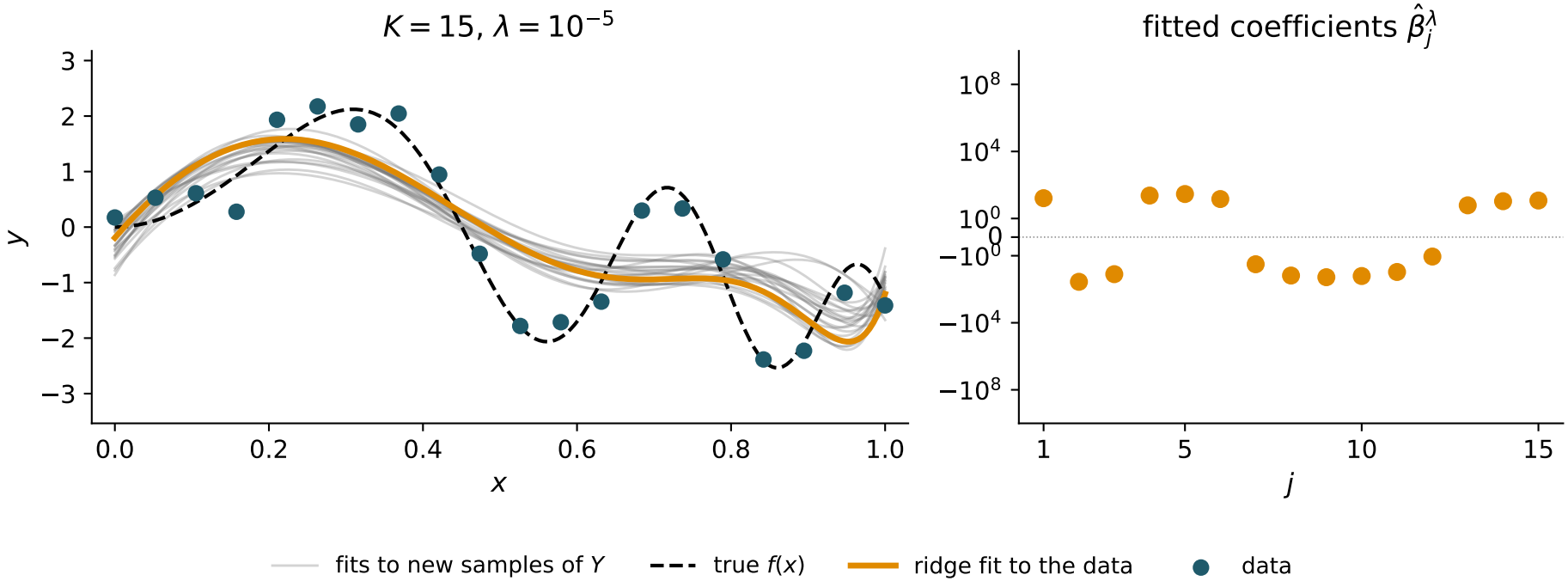
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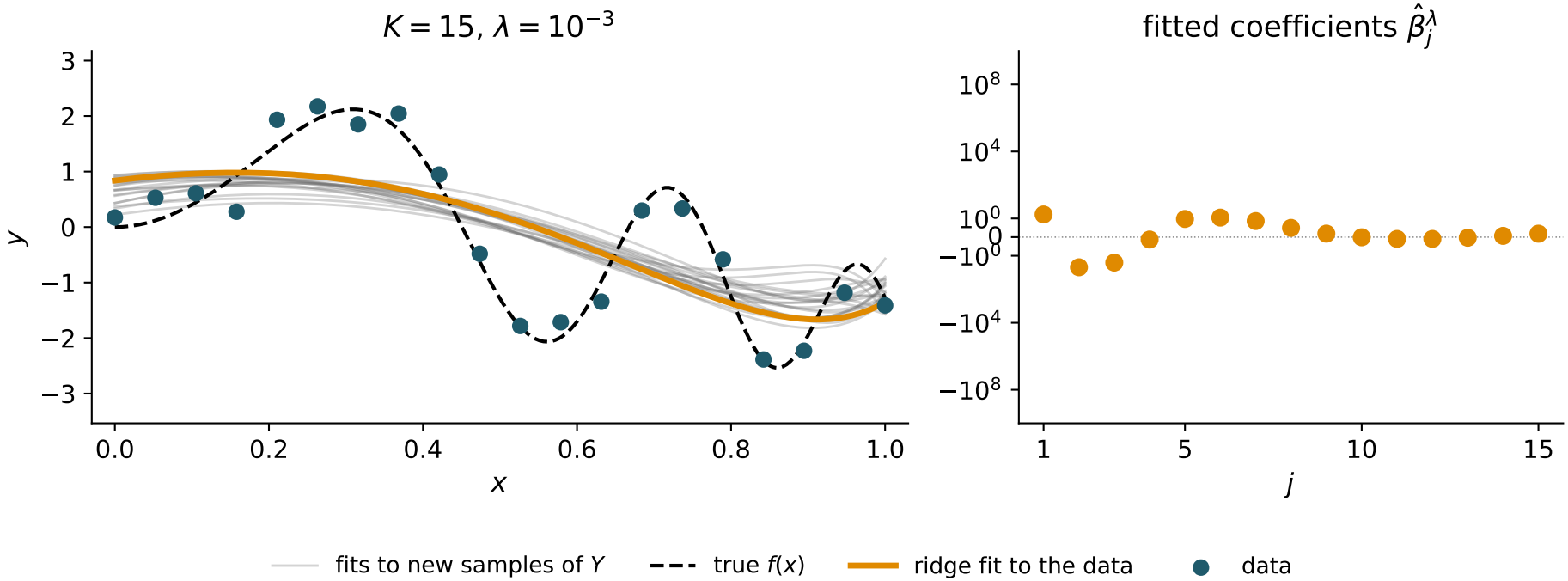
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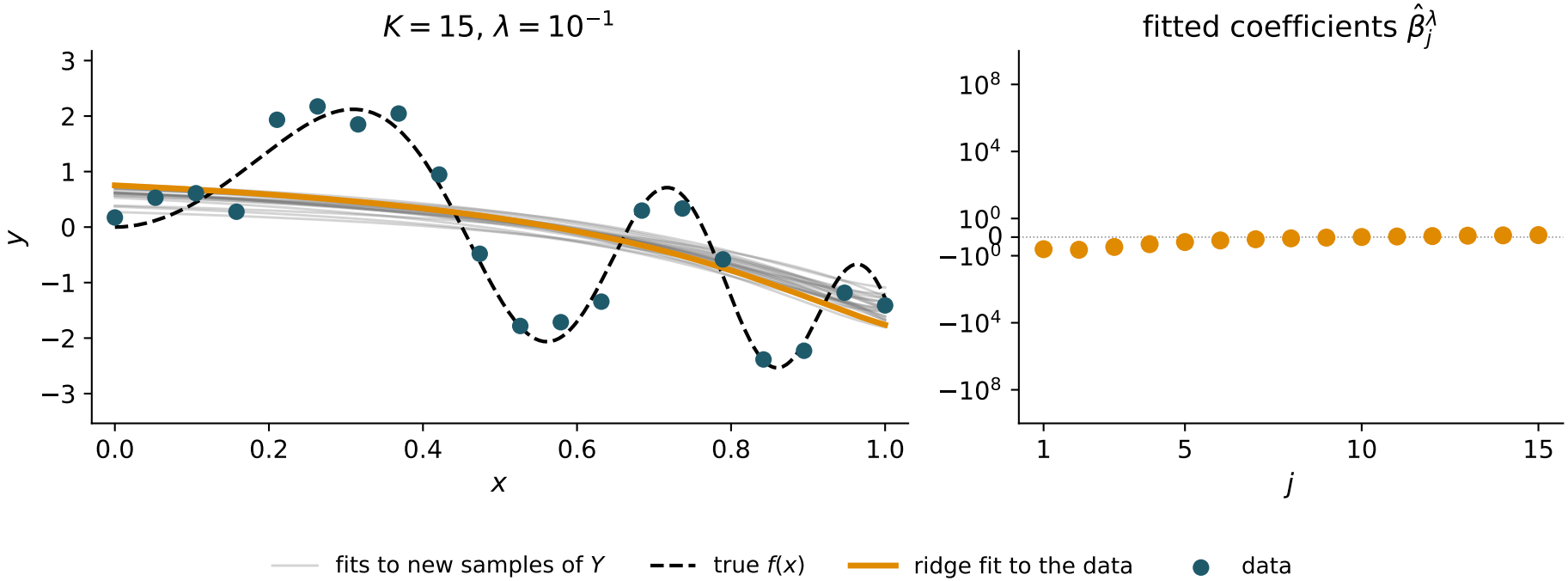
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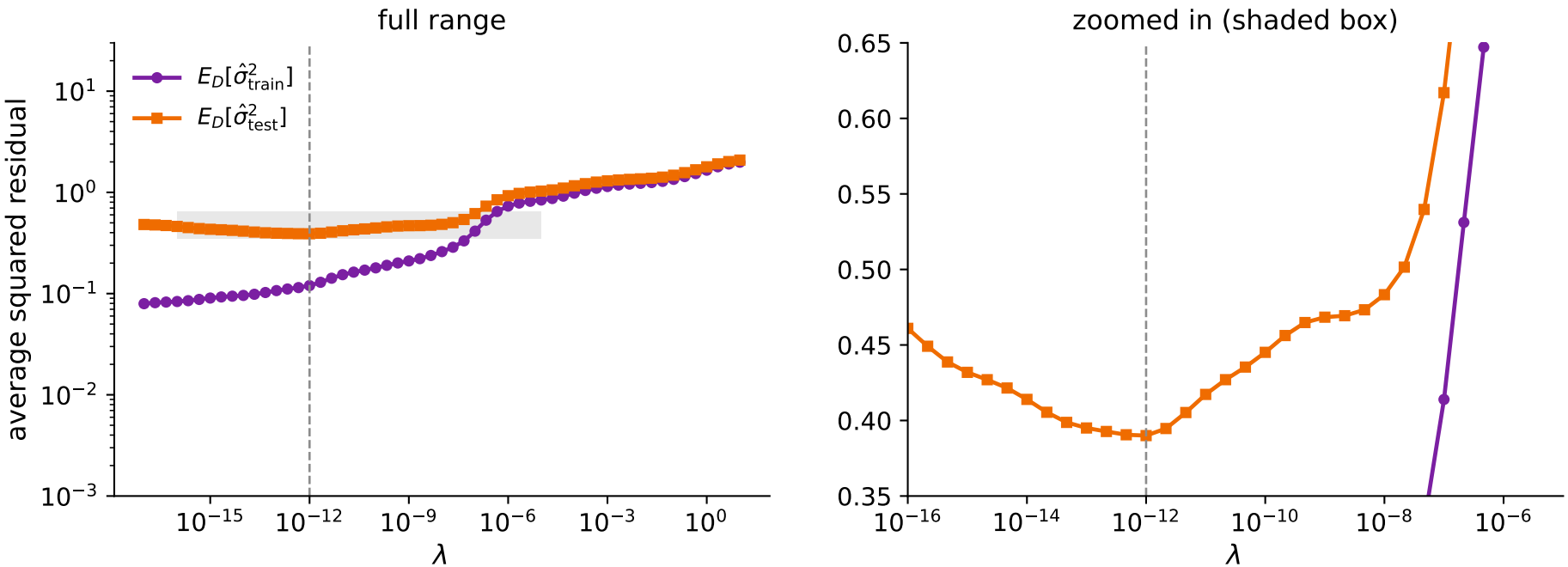


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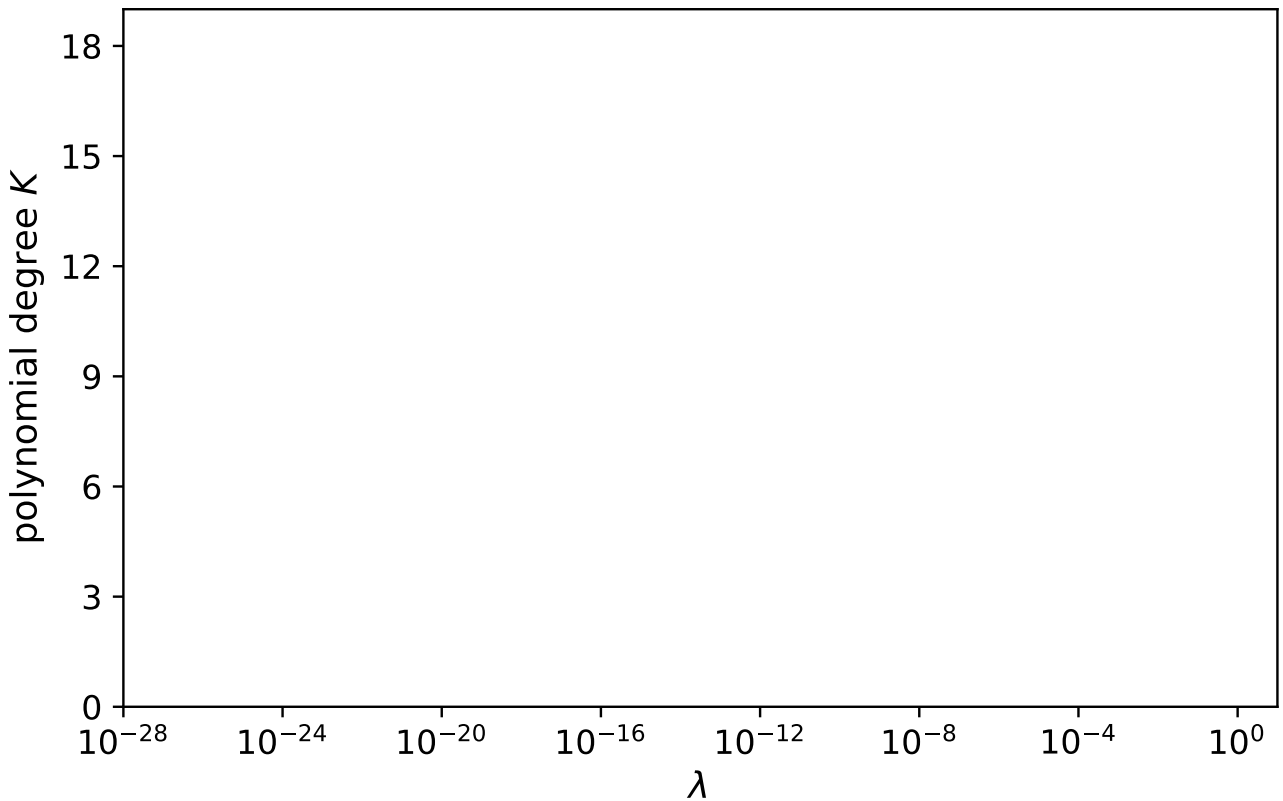
Training and test error vs. λ

Each error is averaged over 400 simulated training sets of the same 20 x values.



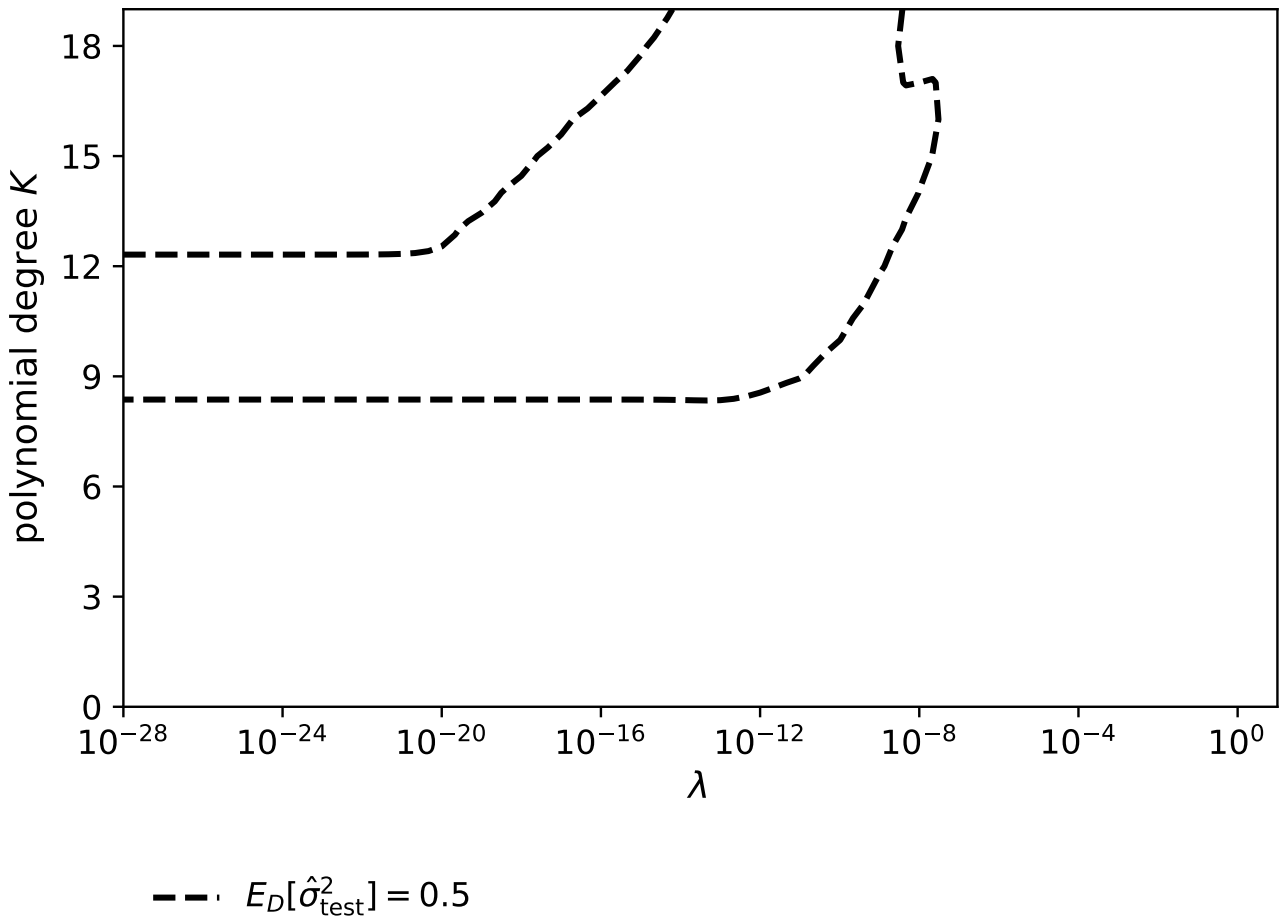
Understanding the error in the K–λ plane

Each point is a choice of degree K and penalty λ. What do the lines where $E_D[\hat{\sigma}^2_{\text{test}}]$ is constant look like?

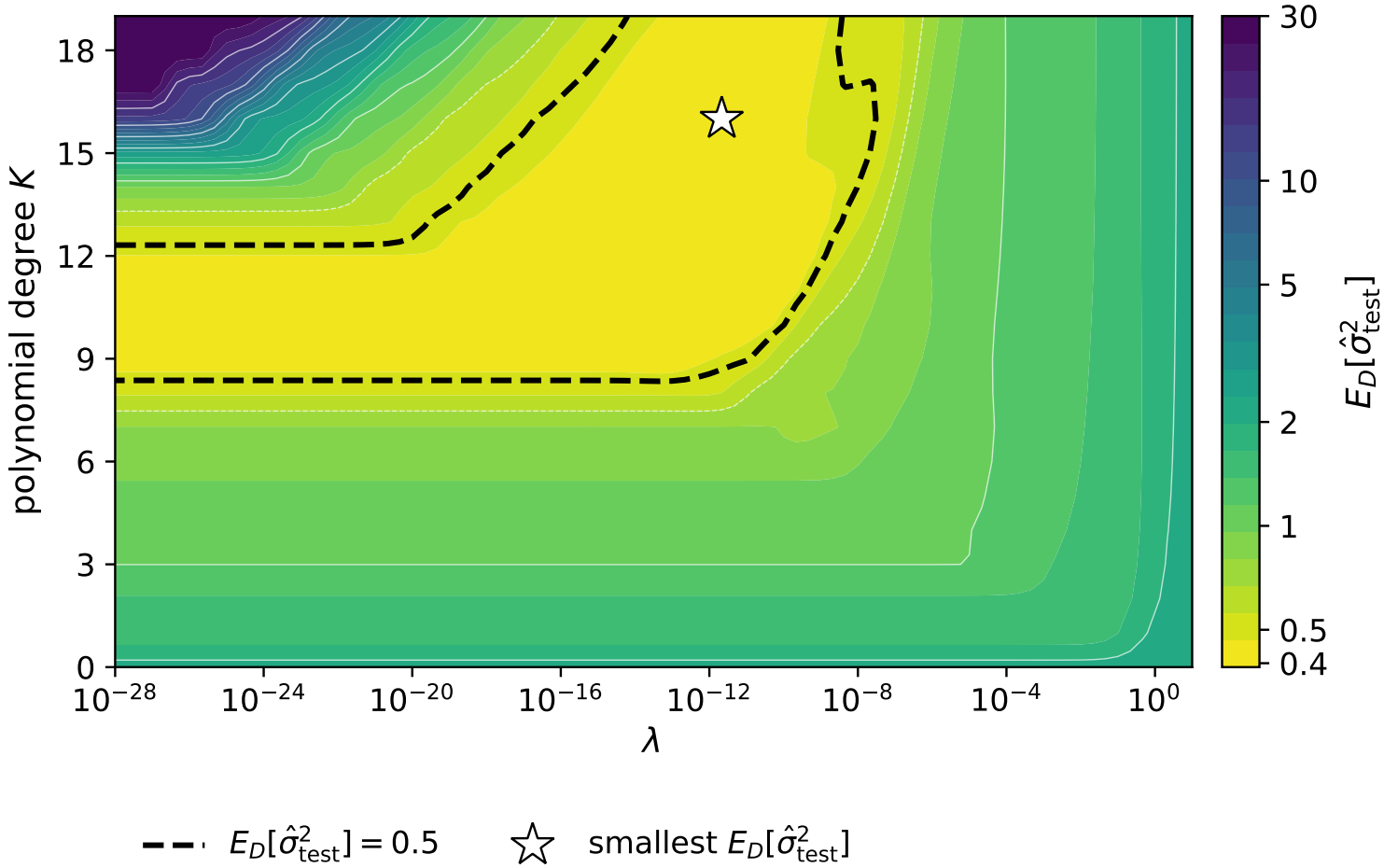


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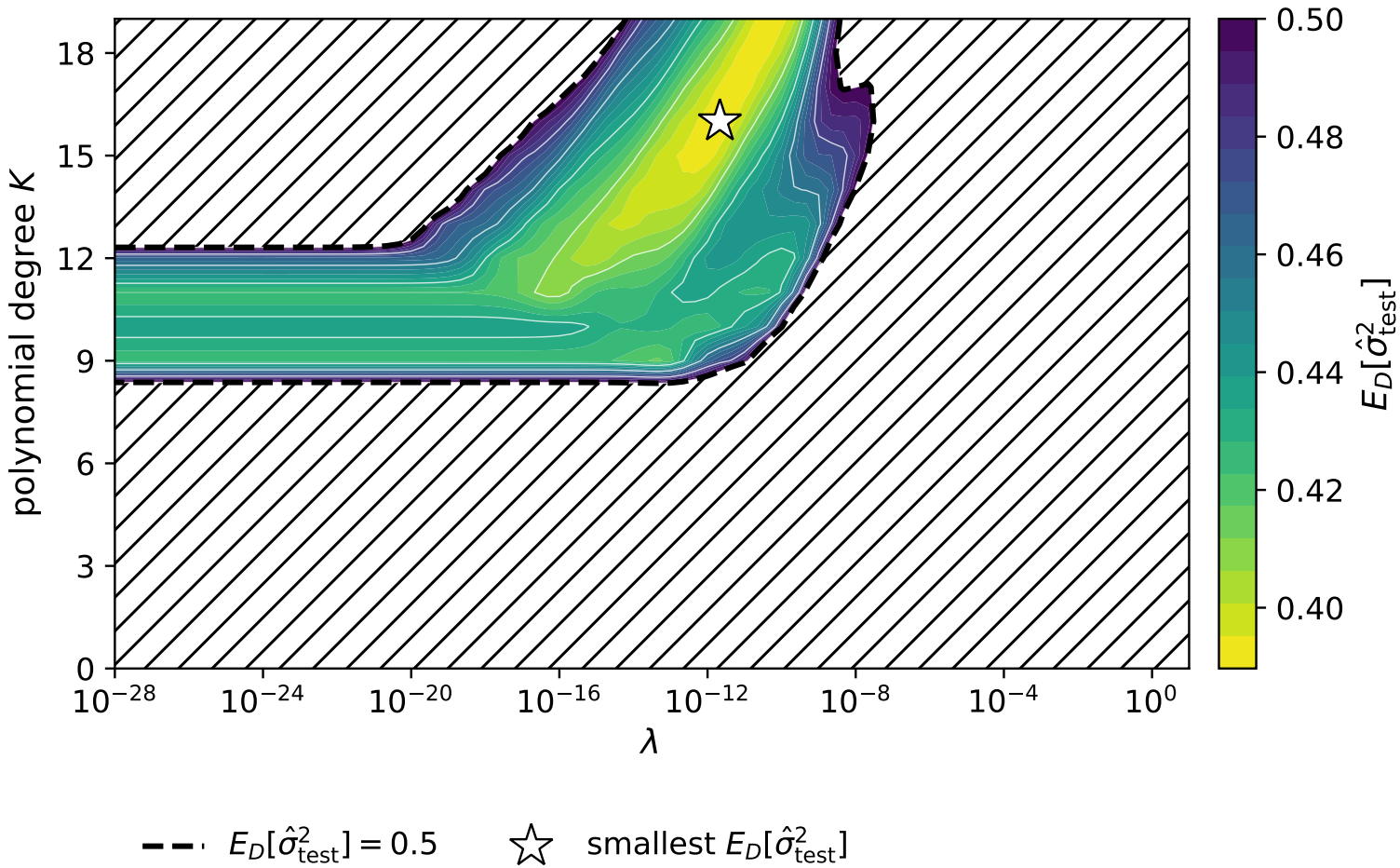
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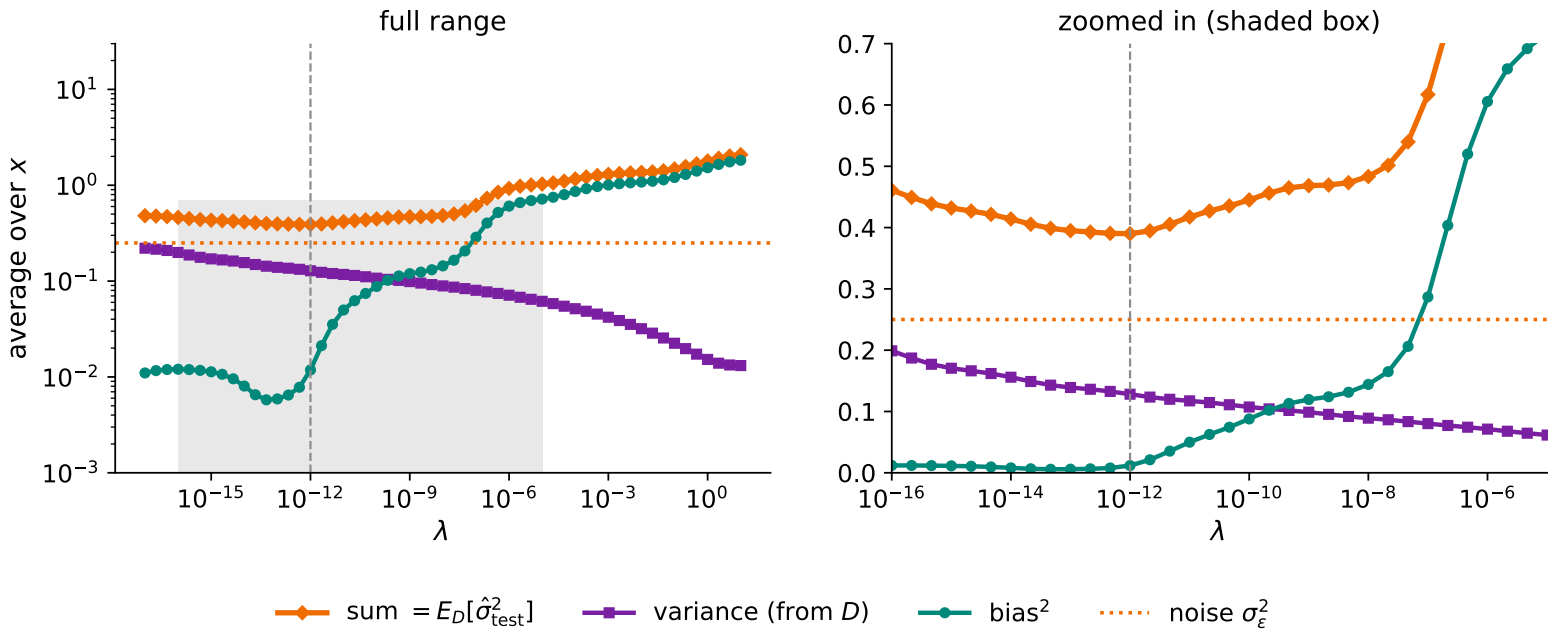
Test error over K and λ



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Bias and variance vs. λ



The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

training set

$N_{\text{train}} = 4000$ variants

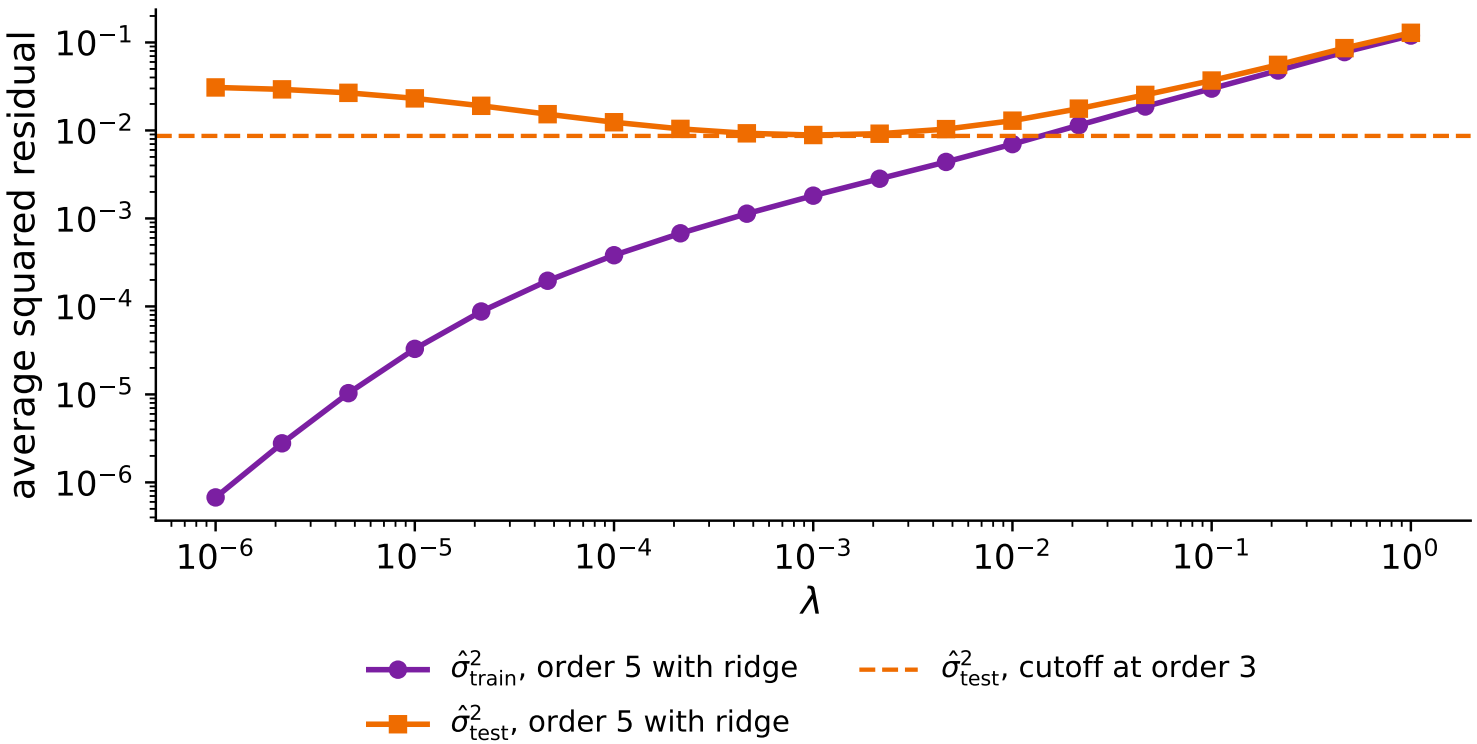
fit with ridge regression

minimize $\quad L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda\left(\sum_i \beta_i^2 + \sum_{i<k} J_{ik}^2 + \sum_{i<k<l} J_{ikl}^2 + \cdots\right)$

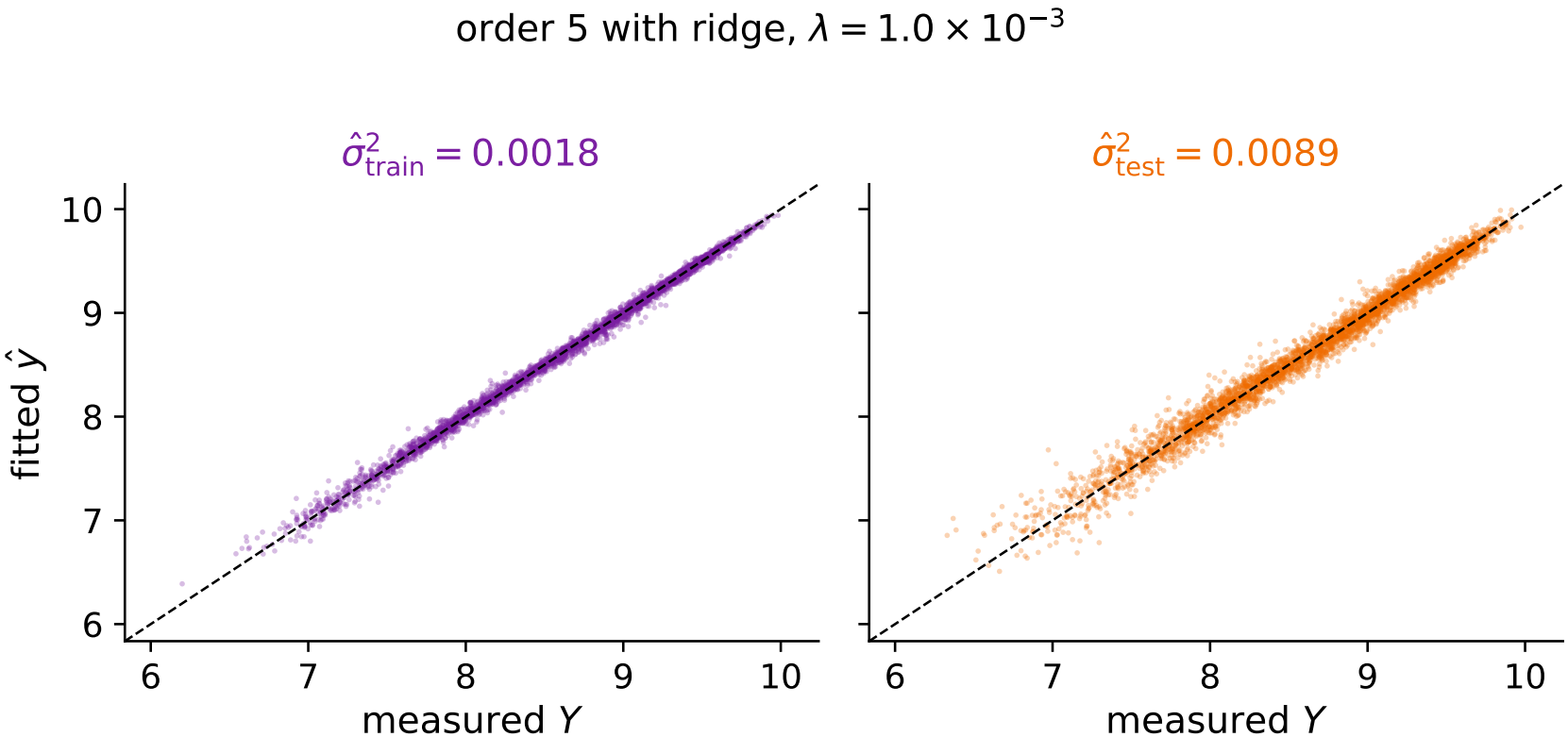
every coefficient except β_0 gets the same penalty

Regularization for the spike protein

All interactions up to order 5 (4944 coefficients), fit with ridge to the same 4000 training variants as on 11/6 and scored on the same 4000 test variants. The dashed line is least squares with the cutoff at order 3.



Fitted vs. measured binding with ridge



Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$$

$k, l = 0, \ldots, 159$: one feature for each pixel

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

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representation in feature space

$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$

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probability model

$$Y = \sum_{k,l} \beta_{k,l}\, \phi_{k,l}(x, y) + \epsilon$$

Regularization for images: setup

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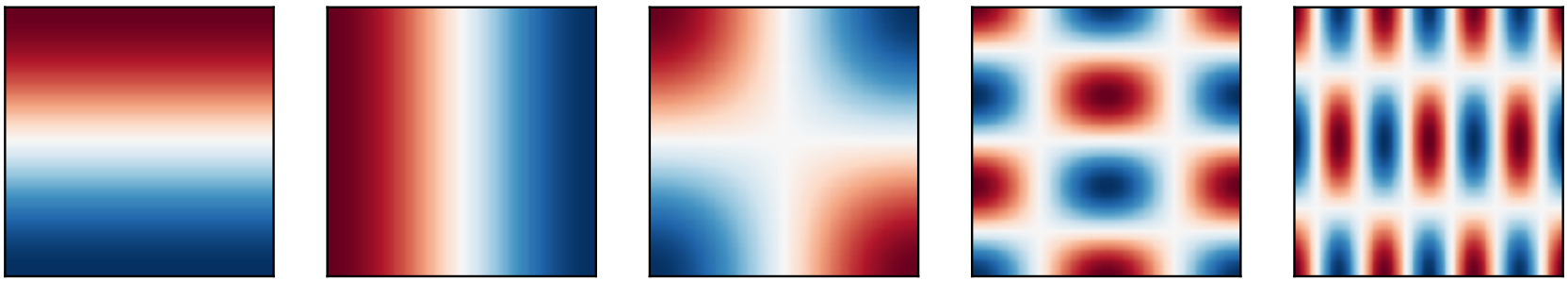
regularization

$$L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \sum_{k,l} (k^2 + l^2)\, \beta^2_{k,l}$$

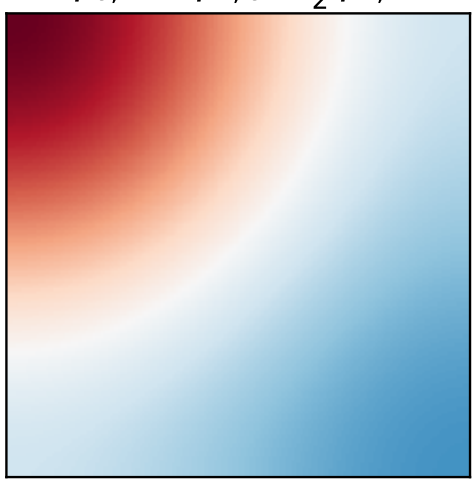
high frequencies cost more

The Fourier features

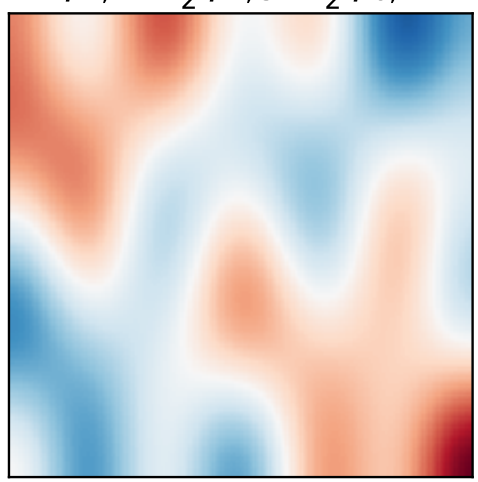
$\phi_{0,1}$ $\phi_{1,0}$ $\phi_{1,1}$ $\phi_{2,3}$ $\phi_{6,2}$



$\phi_{0,1} + \phi_{1,0} + \frac{1}{2}\phi_{1,1}$

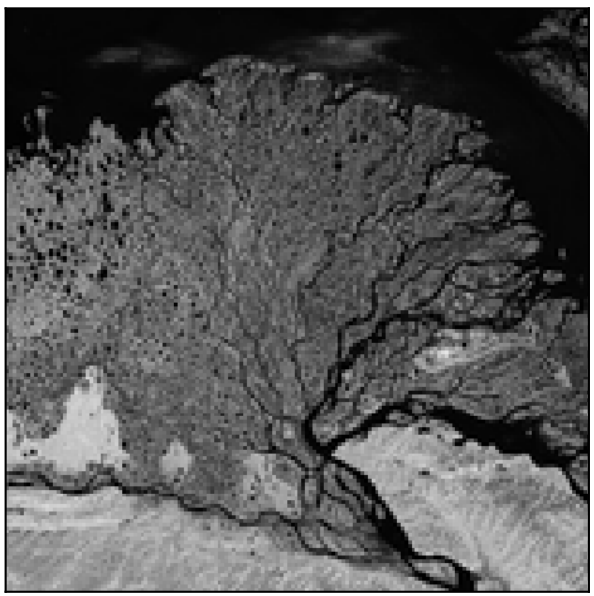


$\phi_{1,1} - \frac{1}{2}\phi_{2,3} + \frac{1}{2}\phi_{6,2}$

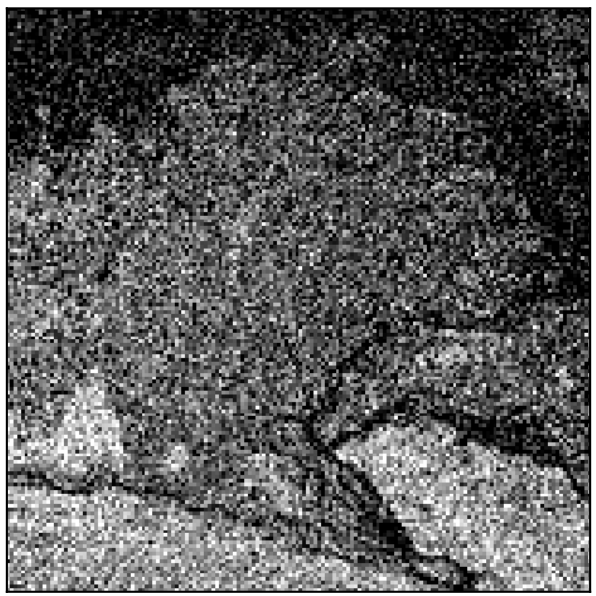


Smoothing a noisy image

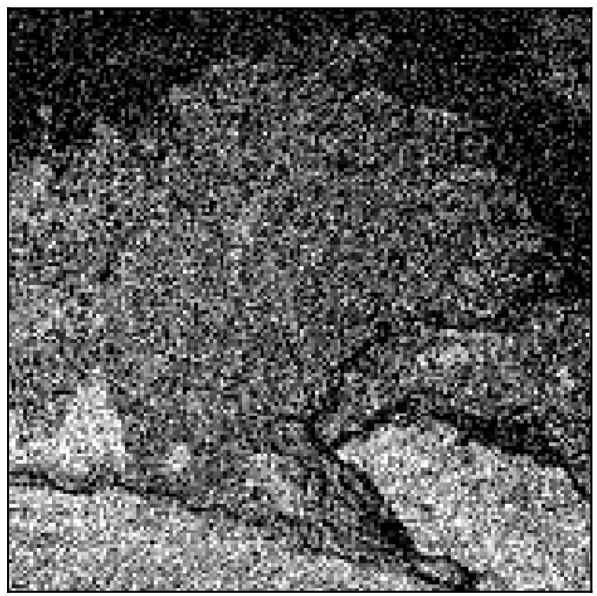
clean image (unknown)



noisy image (data)



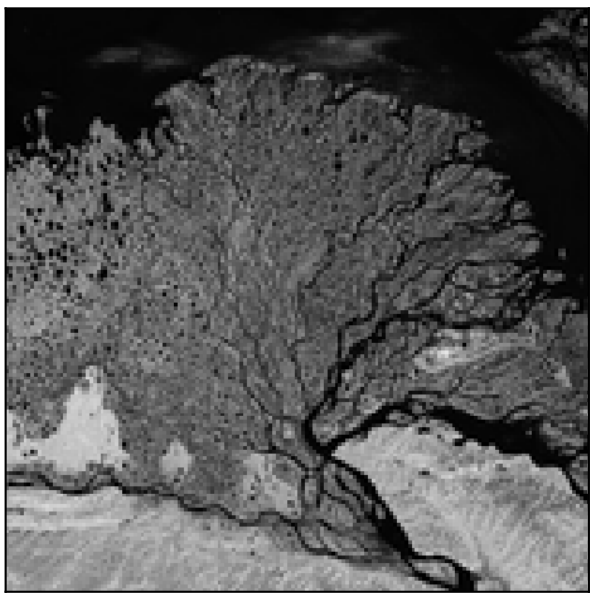
fit, $\lambda = 0$ (least squares)



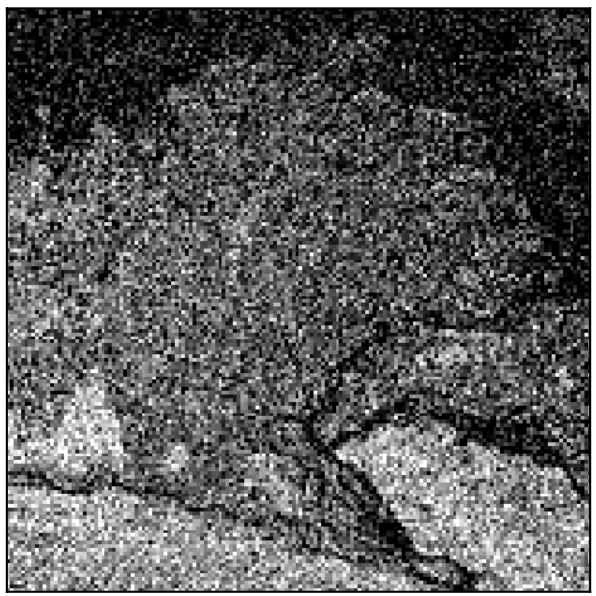
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

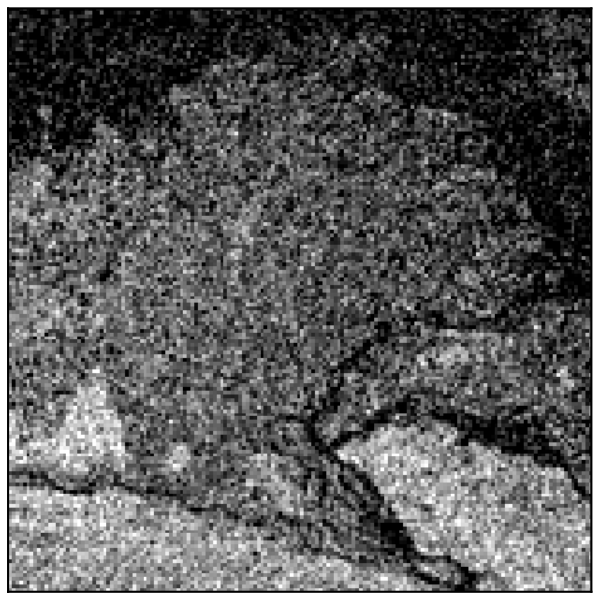
clean image (unknown)



noisy image (data)



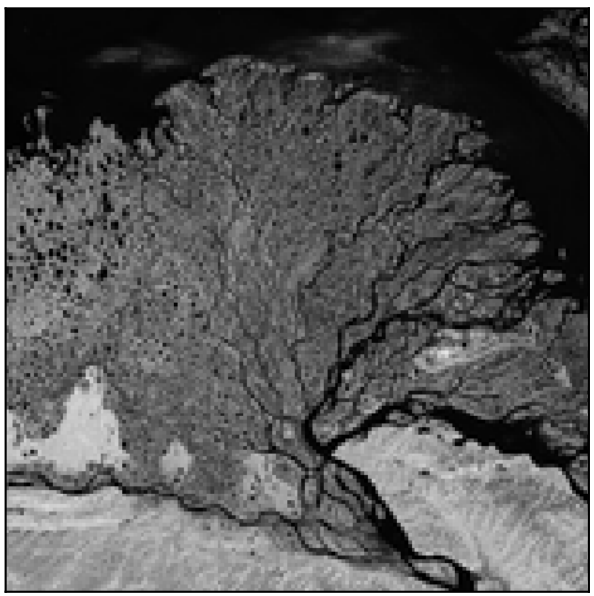
fit, $\lambda = 10^{-9}$



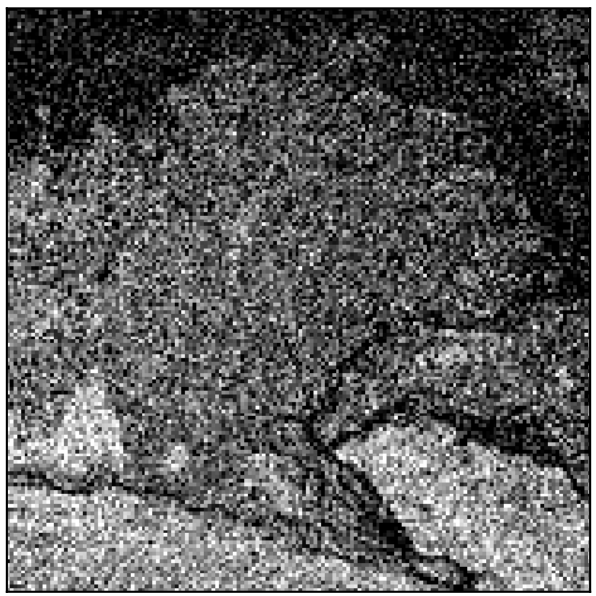
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

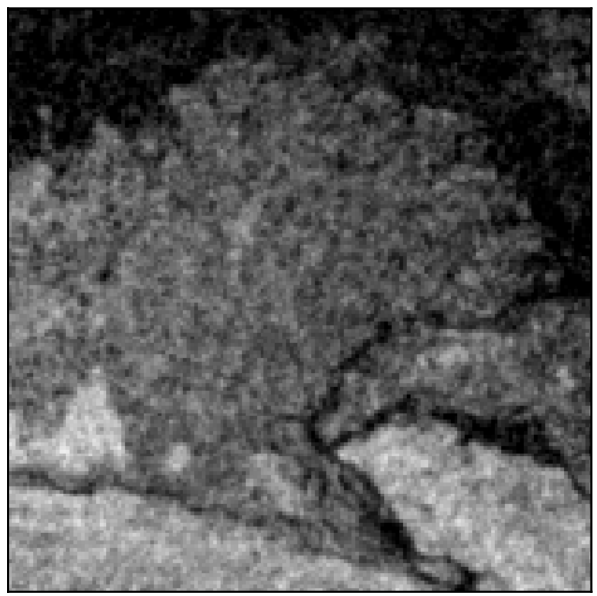
clean image (unknown)



noisy image (data)



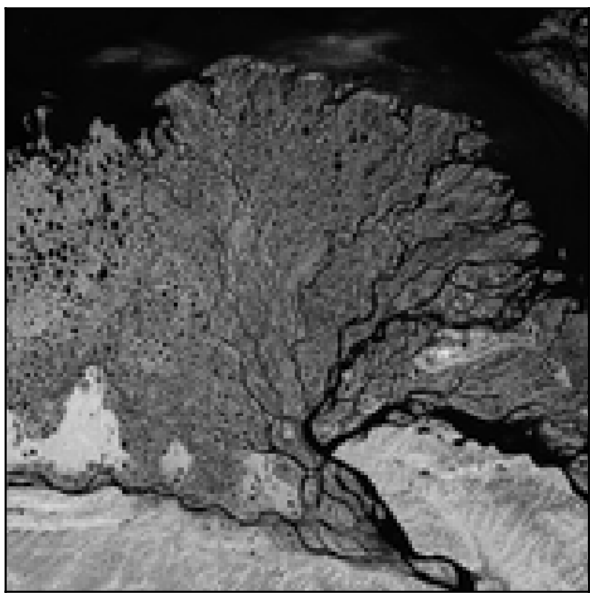
fit, $\lambda = 10^{-8}$



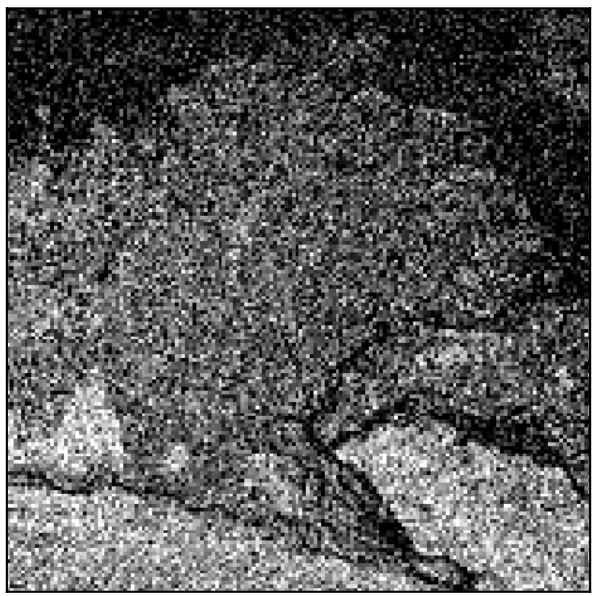
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

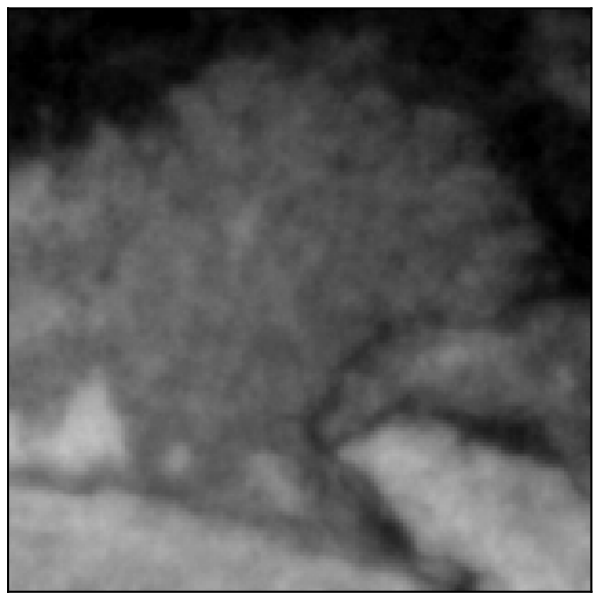
clean image (unknown)



noisy image (data)



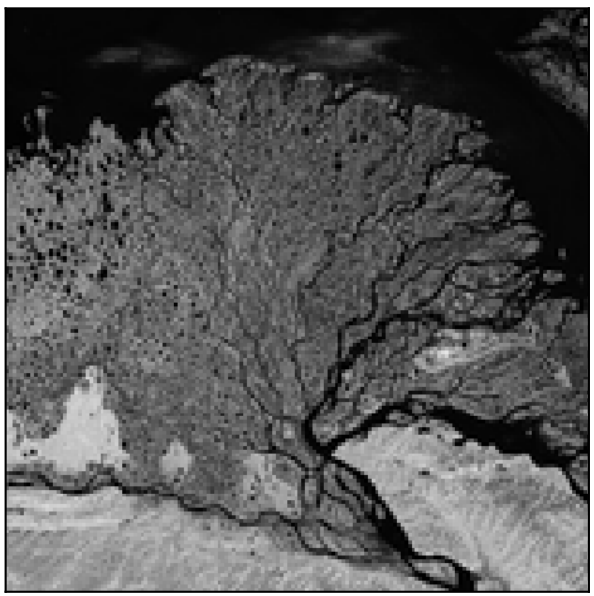
fit, $\lambda = 10^{-7}$



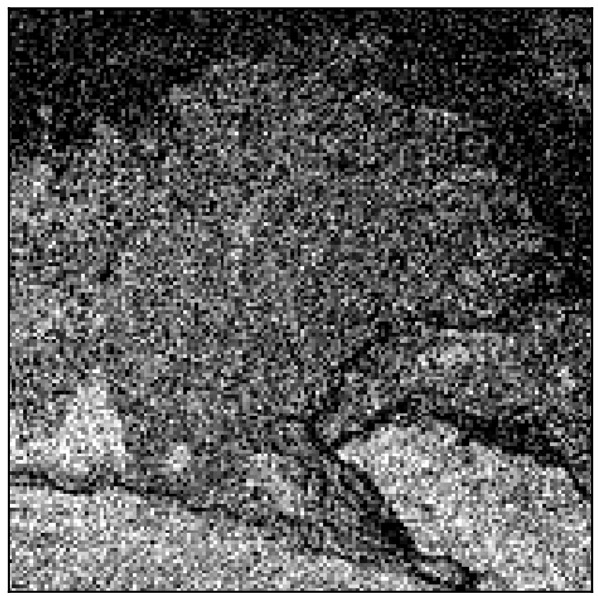
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

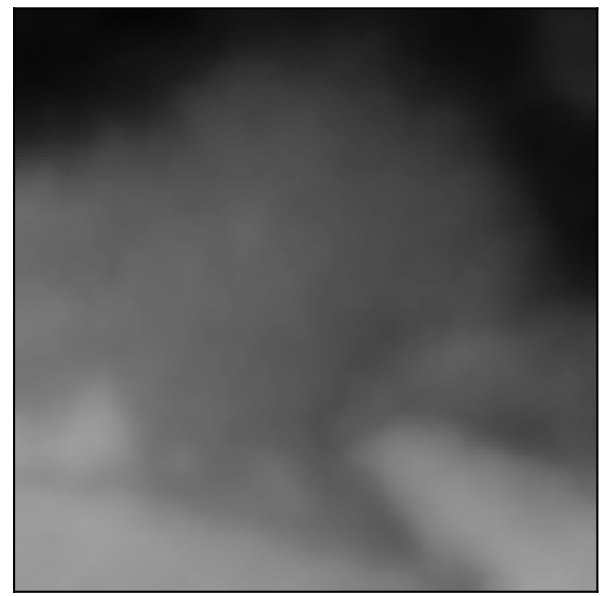
clean image (unknown)



noisy image (data)



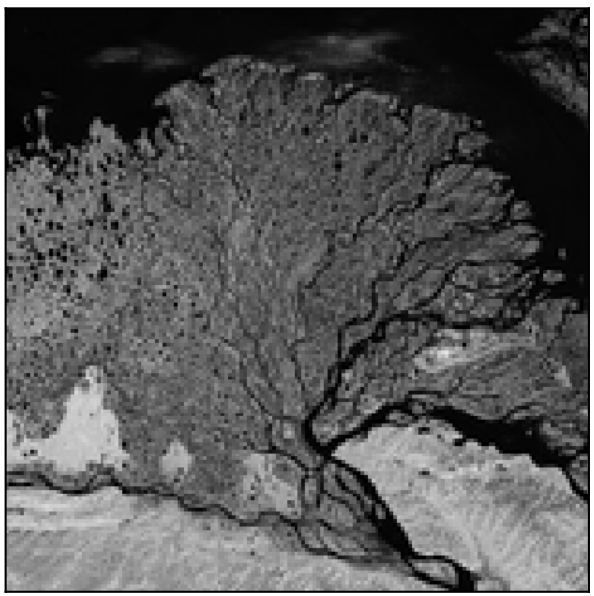
fit, $\lambda = 10^{-6}$



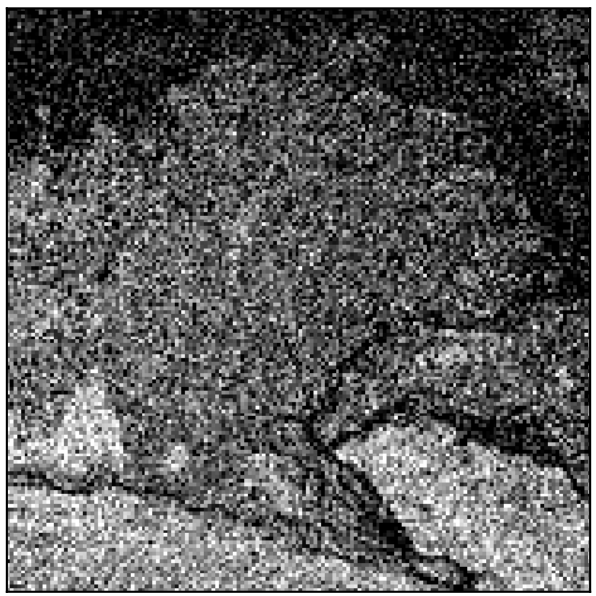
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

clean image (unknown)



noisy image (data)

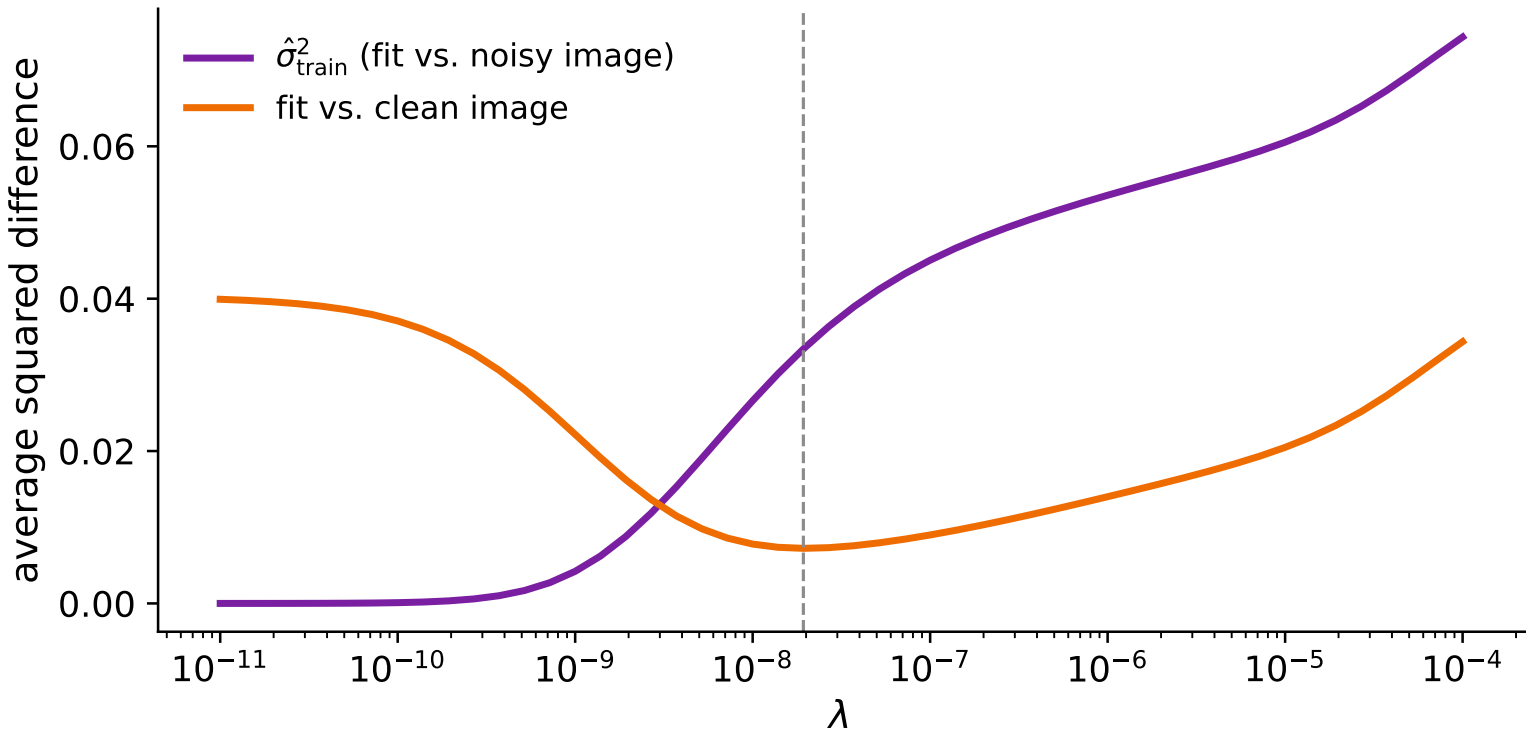


fit, $\lambda = 10^{-5}$



Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Choosing λ for the image



Solutions

1) Car insurance. A biased model prices some groups of drivers too high or too low on average, so those drivers are systematically overcharged or undercharged and the company loses money on the groups it underprices. A model with high variance gives prices that change a lot with the particular data used to fit it, so two similar drivers, or the same driver from one year to the next, can be quoted very different prices, and the company's revenue is less predictable.

2) Cancer treatment. Without drug A the options are no drugs (0), C alone ($\beta_C = 2$), B alone ($\beta_B = 3$), and B and C ($\beta_B + \beta_C + J_{BC} = 3 + 2 - 4 = 1$). The best is drug B alone, with an expected reduction of 3 mm. Adding C to B makes it worse because of the interaction $J_{BC} = -4$.

3) $R^2 \approx 1 - \hat{\sigma}^2_{\text{train}} / \hat{\sigma}^2_Y = 1 - 0.5/2 = 0.75$.

4) (a) $\hat{\sigma}^2_{\text{train}}$ decreases (it can never increase, since the larger model can always set the new coefficient to 0) and reaches 0 once the polynomial passes through every training point. $\hat{\sigma}^2_{\text{test}}$ first decreases and then increases, the U shape, and becomes very large as K approaches N_{train}.
   (b) No. The variance term $E_X\left[\text{var}_D(\hat{y}(X, D))\right]$ is an average over X. A large average does not mean the standard error of $\hat{y}(x, D^{\text{train}})$ is large at every x.
   (c) No. $E_D[\hat{\sigma}^2_{\text{test}}] = E_X\left[\text{MSE}_{\hat{y}(X,D)}\right] + \sigma^2_\epsilon \geq \sigma^2_\epsilon$.